CHAPTER 8

MECHANICAL PROPERTIES OF SOLIDS

CHAPTER ESSENCE AT A GLANCE:
Solid matter acts like a microscopic lattice of springs! Stretch a wire and it resists (stress); when stress is small, strain is directly proportional (Hooke's Law). The slope of the stress-strain curve is Young's Modulus—the ultimate stiffness metric!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Stress and Strain:** Stress = Restoring Force or Area (N/m^2 or Pa). Strain = Change in dimension or Original dimension (dimensionless ratio).

• **Hooke's Law & Elastic Moduli:** Within elastic limit: Stress = Modulus × Strain. Young's Modulus Y = (F L) divided by (A ΔL); Shear Modulus G = (F divided by A) divided by θ; Bulk Modulus B = - ΔP divided by (ΔV divided by V).

• **Compressibility:** Compressibility k is the reciprocal of Bulk modulus: k = 1 divided by B. For water, B = 2.0×10^9 Pa implies k = 5.0×10^{-10} Pa^{-1}.

• **Why Steel is More Elastic Than Rubber:** In physics, 'more elastic' means requiring greater stress for a given strain. Y steel > Y rubber, so steel is more elastic!

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Stress} = \frac{F}{A}, \quad \text{Strain} = \frac{\Delta L}{L}$$

Tensile Stress and Longitudinal Strain

$$Y = \frac{FL}{A\Delta L}, \quad B = -\frac{P}{\frac{\Delta V}{V}}$$

Young's Modulus and Bulk Modulus

$$G = \frac{F}{A\theta}, \quad \sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

Shear Modulus and Poisson's Ratio

$$u = \frac{1}{2} \times \text{Stress} \times \text{Strain} = \frac{1}{2} Y (\text{Strain})^2$$

Elastic Energy Stored per Unit Volume

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Steel is MORE elastic than rubber! In physics, 'more elastic' means it requires greater stress to produce a given strain.

• **Alert:** The area under the force-extension curve equals the work done in stretching the wire (Stored Energy = ½ F ΔL).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define stress and strain. State their SI units. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Stress]** Stress is restoring force per unit cross-sectional area: Stress = F divided by A (SI unit: N m^{-2} or Pascal). [1 Mark]

• **[Step 2: Strain]** Strain is fractional deformation: Strain = Change in dimension or Original dimension (Dimensionless quantity, no units). [1 Mark]

• **Key Exam Result:** Stress = F divided by A (Pascal), Strain = ΔL divided by L (Dimensionless)

2. Explain why steel is more elastic than rubber scientifically. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Scientific Explanation]** For an identical stretching force applied to equal length and cross-section wires of steel and rubber, steel produces far smaller strain (Δ L steel << Δ L rubber). Since Young's Modulus Y = Stress or Strain, a smaller strain for identical stress means Y steel >> Y rubber. Higher Young's modulus means greater resistance to deformation, so steel is more elastic. [2 Marks]

• **Key Exam Result:** Y steel >> Y rubber; steel offers greater restoring stress

3. Define: (a) Young's Modulus (Y) (b) Shear Modulus (G) (c) Bulk Modulus (B). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Young's Modulus]** (a) Ratio of longitudinal stress to longitudinal strain: Y = (F divided by A) divided by (ΔL divided by L) = (F L) divided by (A ΔL). [1 Mark]

• **[Step 2: Shear Modulus]** (b) Ratio of shearing stress to shearing strain (angle of shear θ): G = (F divided by A) divided by θ. [1 Mark]

• **[Step 3: Bulk Modulus]** (c) Ratio of hydraulic stress to volumetric strain: B = - p divided by (ΔV divided by V). [1 Mark]

• **Key Exam Result:** Y = (FL) divided by (A ΔL), G = (F divided by A) divided by θ, B = -p divided by (ΔV divided by V)

4. Differentiate between ductile materials and brittle materials with examples and stress-strain curve characteristics. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Ductile Materials]** Materials that have a large plastic deformation range between elastic limit and fracture point. They can be drawn into thin wires (e.g. Copper, Aluminium, Gold). [1.5 Marks]

• **[Step 2: Brittle Materials]** Materials in which the breaking fracture point is very close to the elastic limit with negligible plastic deformation. They snap suddenly under stress (e.g. Glass, Cast iron). [1.5 Marks]

• **Key Exam Result:** Ductile: wide plastic zone (copper); Brittle: snaps near elastic limit (glass)

5. Derive an expression for the elastic potential energy stored per unit volume (energy density) of a stretched metallic wire. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Work Done in Stretching]** When a wire of length L and area A is stretched by length x, restoring force F = (Y A divided by L) x. Work done for stretching dx: dW = F dx = (Y A divided by L) x dx. [1 Mark]

• **[Step 2: Total Work Done]** W = $\int_0^{\Delta L}$ (Y A divided by L) x dx = half (0.5) (Y A divided by L) $(\Delta L)^2$ = half (0.5) x [(Y A ΔL) divided by L] x ΔL = half (0.5) x F x ΔL. [1 Mark]

• **[Step 3: Energy Density]** Energy density u = Work or Volume = (half (0.5) x F x ΔL) divided by (A x L) = half (0.5) x (F divided by A) x (ΔL divided by L) = half (0.5) x Stress x Strain. [1 Mark]

• **Key Exam Result:** Energy Density u = half (0.5) x Stress x Strain = half (0.5) Y x $(Strain)^2$

6. A steel wire of length 4.7 m and cross-section 3.0×10^{-5} m^2 stretches by the same amount as a copper wire of length 3.5 m and cross-section 4.0×10^{-5} m^2 under a given load. Find the ratio of Young's modulus of steel to that of copper. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Young's Modulus Formula]** Y = (F L) divided by (A ΔL). Since load F and elongation ΔL are identical: Y proportional to (L divided by A). [1 Mark]

• **[Step 2: Ratio Setup]** Y s divided by Y c = (L s divided by A s) divided by (L c divided by A c) = (L s divided by L c) x (A c divided by A s). [1 Mark]

• **[Step 3: Numerical Evaluation]** Y s divided by Y c = (4.7 divided by 3.5) x (4.0×10^{-5} divided by 3.0×10^{-5}) = (4.7 × 4) divided by (3.5 × 3) = 18.8 divided by 10.5 ≈ 1.79. [1 Mark]

• **Key Exam Result:** Y steel divided by Y copper = 1.79 : 1

7. Draw a typical stress-strain curve for a metallic wire under increasing tensile load. Clearly mark and explain: (a) Proportional limit (b) Elastic limit (yield point) (c) Ultimate tensile strength (d) Fracture (breaking) point. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Curve Description & Region OA]** (a) OA (Proportional limit): Stress is linearly proportional to strain; Hooke's law is obeyed strictly. [1 Mark]

• **[Step 2: Region AB & Point B]** (b) Point B (Elastic limit / Yield point): The maximum stress up to which the wire returns to its original length when unloading. Beyond B, permanent set occurs. [1 Mark]

• **[Step 3: Point D (Ultimate Tensile Strength)]** (c) Point D: The highest point of stress the material can endure before necking and thinning begins. [1 Mark]

• **[Step 4: Point E (Fracture Point)]** (d) Point E (Breaking point): The point where the wire snaps. If E is far from D, material is ductile; if close, material is brittle. [1 Mark]

• **Key Exam Result:** OA (Hooke's law) -> B (Elastic limit) -> D (UTS) -> E (Fracture)